

Complex Python

Lesson #5

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Error Handling

In programming, errors or bugs will happen. Error handling lets you code for potential problems and decide how to handle them without it crashing. Without error handling, if your program encounters a situation it doesn't know how to handle (like a user dividing by zero or trying to open a file that doesn't exist), the entire script will stop abruptly, often displaying a confusing error message to the user. Good error handling allows your program to "catch" these issues, log them, or provide a user-friendly message, allowing the script to continue running safely.

try Block

try block: You put the code that might cause an error here. This is where you place the code that you suspect might fail. Python will attempt to execute this code; if an error occurs, it immediately stops and jumps to the except block.

```
try:
    number = int(input("Enter a number to divide 10 by: "))
    result = 10 / number
    print(f"The result is {result}")
```

except Block

except block: This code only runs if an error occurs inside the try block. This is where you can print a friendly message or log the issue. This is your "rescue" zone. If the code inside the try block crashes, the program execution moves here, where you can define how to respond to specific types of errors, like printing a warning.

```
try:
    number = int(input("Enter a number to divide 10 by: "))
    result = 10 / number
    print(f"The result is {result}")
except ZeroDivisionError:
    print("Error: You can't divide by zero!")
```

finally Block

finally block: This code runs no matter what, whether an error occurred or not. It is typically used for "cleanup" tasks, like closing a file or disconnecting from a database. This is an optional block that runs no matter what, regardless of whether an error occurred or not. It is typically used for cleanup tasks, such as closing a file or disconnecting from a database.

```
try:
    number = int(input("Enter a number to divide 10 by: "))
    result = 10 / number
    print(f"The result is {result}")
except ZeroDivisionError:
    print("Error: You can't divide by zero!")
finally:
    print("Done")
```



Activity Time!

```python
try:
    number = int(input("Enter a number to divide 10 by: "))
    result = 10 / number
    print(f"The result is {result}")
except ZeroDivisionError:
    print("Error: You can't divide by zero!")
except ValueError:
    print("Error: Please enter a valid number.")
finally:
    print("Done")
```

```
Enter a number to divide 10 by: 0
Error: You can't divide by zero!
Done
```

Activity Time!

Thank You!

Any Questions?